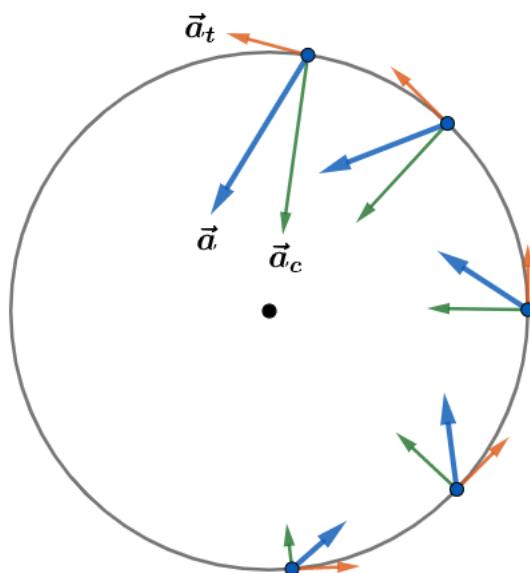


2021 F=ma Exam: Problem 13

Kevin S. Huang

Since the car is undergoing circular motion with increasing speed, it has a tangential acceleration (orange arrows) as well as an increasing centripetal acceleration (green arrows) because $a_c = v^2/r$.



Thus, the total acceleration (blue arrows) is shown in diagram D.